

0 4 . 1

Where does transcription occur in a eukaryotic cell?

[1 mark]

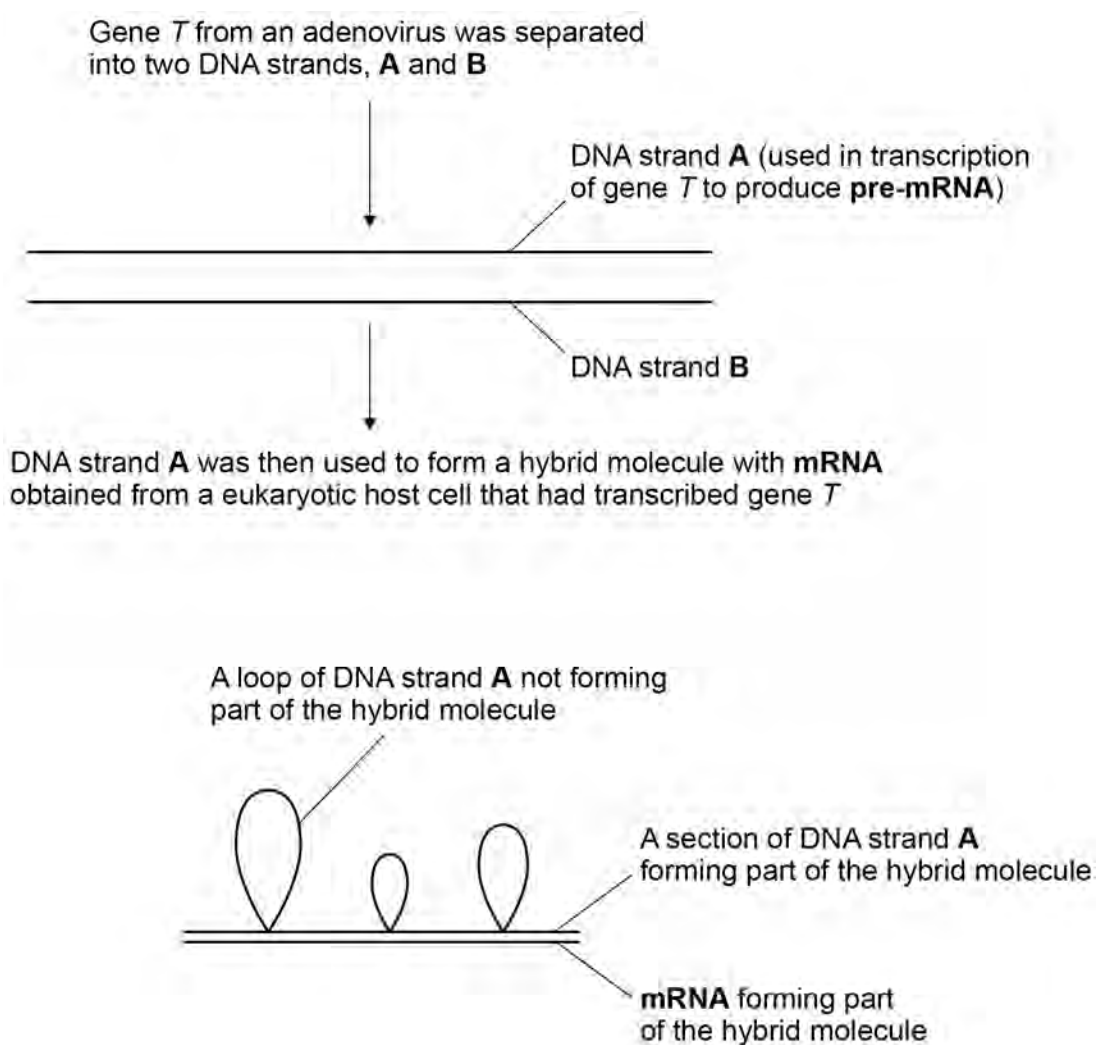
0 4 . 2

The genome of an adenovirus is a single, linear molecule of double-stranded DNA. Adenoviruses use eukaryotic host cells to transcribe their genes in protein synthesis. The process of transcription of adenovirus genes is similar to the process of transcription of genes in eukaryotes.

Scientists investigated the process of transcription of this viral DNA.

Figure 4 shows one of the experiments carried out by these scientists.

Figure 4



Looking at the results of the experiment in **Figure 4**, the scientists concluded that splicing had taken place.

Use **Figure 4** to describe and explain why the scientists' conclusion was justified.

[3 marks]

0 4 . 3

Describe and explain how the results of the experiment in **Figure 4** would differ if the scientists had used prokaryotic DNA.

[2 marks]

0 4 . 4

Errors in the precise location of splicing in the DNA molecule can lead to mutations.

Explain why.

[1 mark]

7

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